CNN Performance Evaluation System

Presented By

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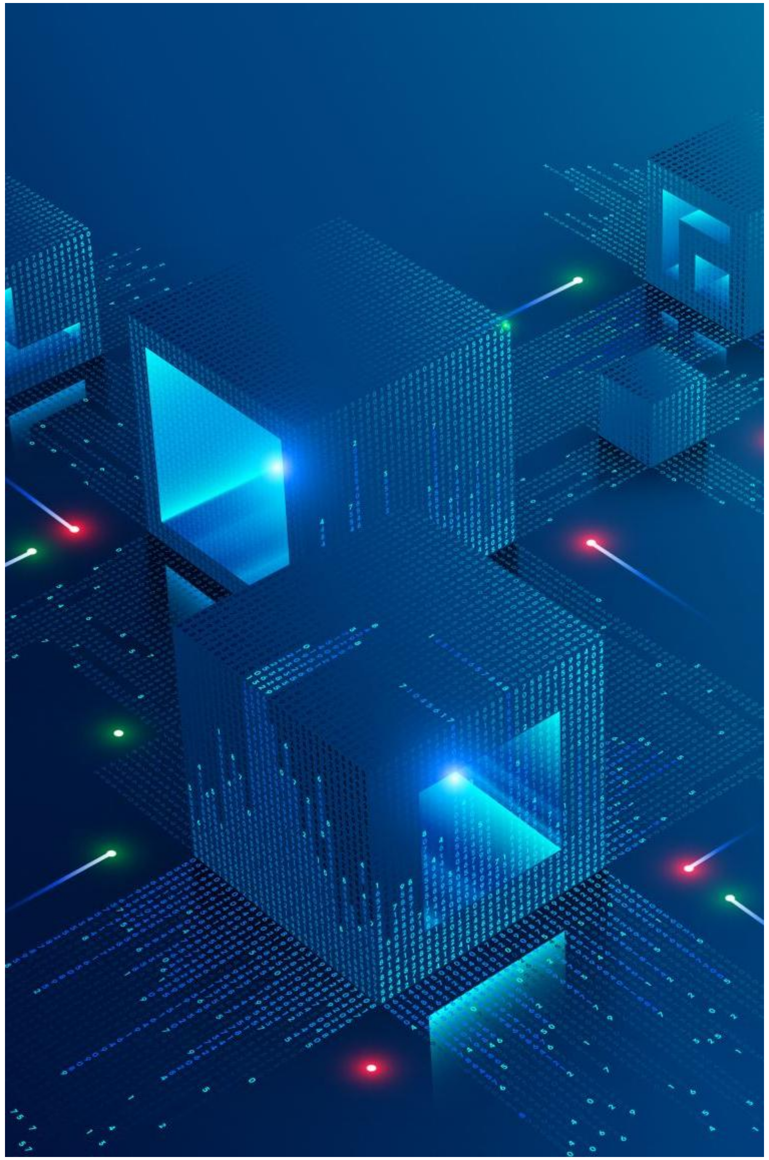
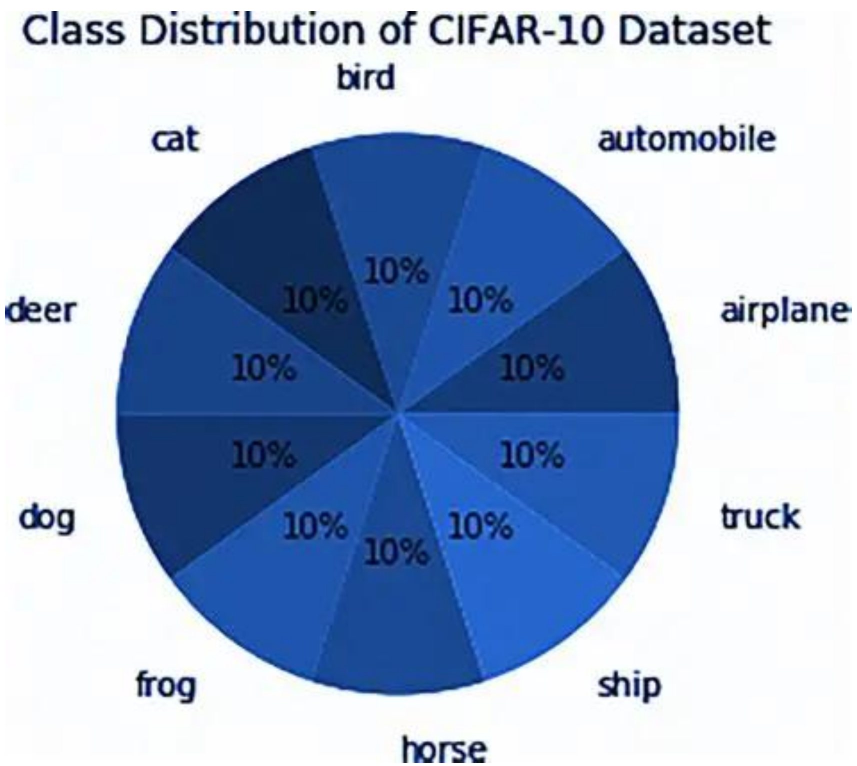
INTRODUCTION

- Convolutional Neural Networks (CNNs) are one of the most important deep learning techniques used in computer vision tasks such as image classification, object detection and facial recognition. CNNs are designed to automatically learn important visual features from images through convolutional operations.
- One major factor that affects CNN performance is the depth of the network. A shallow CNN may only learn basic image features such as edges and textures, while deeper CNNs can learn more complex and abstract patterns. However, increasing network depth also increases computational cost and the risk of overfitting.
- This project investigates how CNN depth influences image classification performance using the CIFAR-10 dataset.



RESEARCH MOTIVATION

- This study investigates how increasing CNN depth influences image classification performance by comparing **Shallow, Medium, and Deep CNN architectures** with respect to:
- Classification Accuracy
- Training Performance
- Confusion Matrix Analysis
- Model Complexity



DATASET STRUCTURE

- The project uses the CIFAR-10 dataset.
- **Dataset Characteristics**
- 60,000 colour images
- 10 image classes
- 32×32 image resolution
- 50,000 training images
- 10,000 testing images

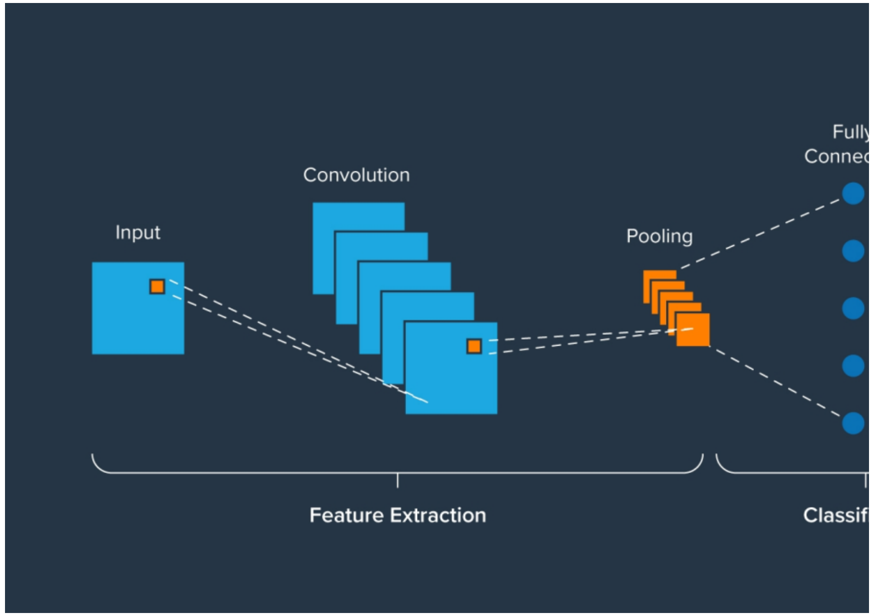
IMPLEMENTATION STEPS

- Data preprocessing and normalization
- Dataset loading using Data Loader
- CNN model creation
- Model training
- Performance evaluation
- Confusion matrix analysis

DATA PROCESSING AND DATASET LOADING

- Converted CIFAR-10 images into PyTorch tensors using ToTensor() so the CNN can process images mathematically.
- Normalized pixel values from **0–255** to approximately **-1 to +1** to improve training stability and learning speed.
- Used DataLoader for efficient batch training and testing to handle large datasets more effectively.
- Batch size set to **64** to balance memory usage and training performance.
- Training data shuffled to improve model learning and prevent overfitting to data order patterns.

CNN MODEL CREATION



- Created three CNN architectures: **Shallow CNN, Medium CNN, and Deep CNN** to compare the effect of network depth on image classification performance.
- The **Shallow CNN** uses **2 convolutional layers** to learn basic visual features, the **Medium CNN** uses **4 convolutional layers** to capture more detailed patterns, while the **Deep CNN** uses **6 convolutional layers and Dropout regularization** to learn complex hierarchical features and improve generalization.
- Used **Convolutional Layers** to automatically extract important image features such as edges, textures, and shapes.
- Applied **ReLU activation functions** to introduce non-linearity and help the model learn complex patterns.
- Used **MaxPooling layers** to reduce image dimensions and improve computational efficiency.
- Added **Fully Connected layers** to perform final image classification into 10 CIFAR-10 classes.

MODEL TRAINING

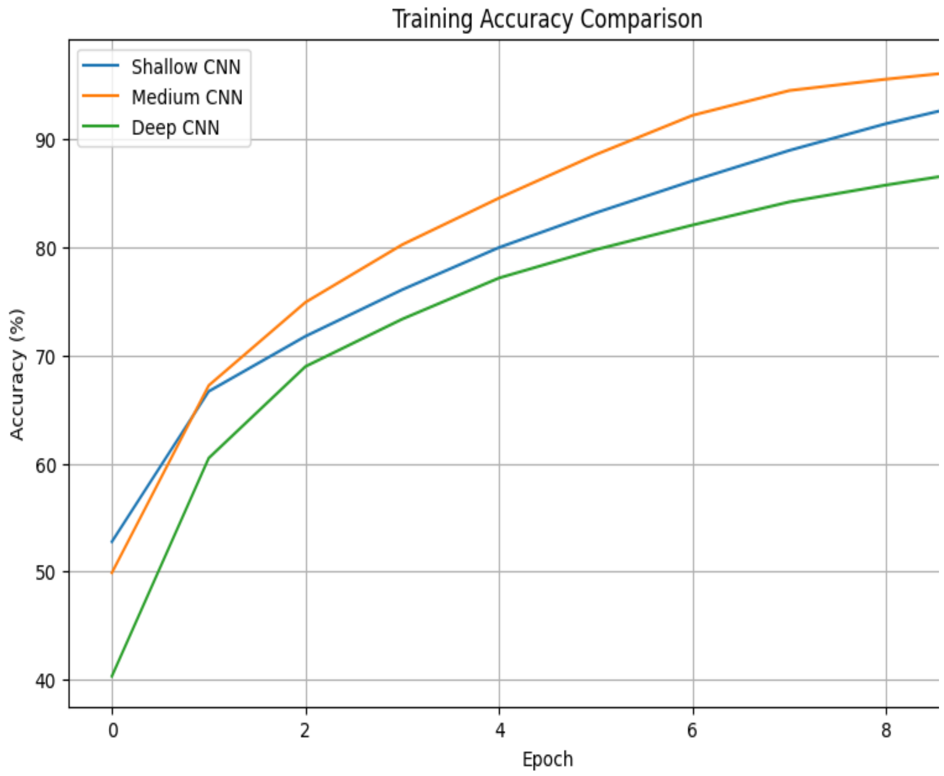
- **Adam Optimizer** was employed to efficiently update network parameters and accelerate convergence during training.
- **Cross-Entropy Loss** was used to quantify classification errors and guide the optimization process for multi-class image recognition.
- **Ten training epochs** were performed to enable the models to progressively learn hierarchical image features from the dataset.
- **Forward propagation** was used to generate class predictions from input images at each training iteration.
- **Backpropagation** was applied to compute gradients and optimize network weights by minimizing prediction errors.
- **Training accuracy and loss metrics** were monitored throughout training to evaluate learning effectiveness and model convergence.

Training Accuracy Comparison

- All CNN models showed a steady increase in training accuracy, indicating successful learning and convergence.
- The **Medium CNN achieved the highest training accuracy (96.49%)**, as it provided the optimal balance between network complexity and feature learning capability.
- The **Shallow CNN achieved 93.58%**, but its limited depth restricted its ability to learn complex image features.
- The **Deep CNN achieved 87.18%**, as its higher complexity and dropout regularization resulted in slower convergence during training.

Key Finding

- The Medium CNN demonstrated the most effective learning of the CIFAR-10 training data, achieving the highest training **accuracy among all models.**

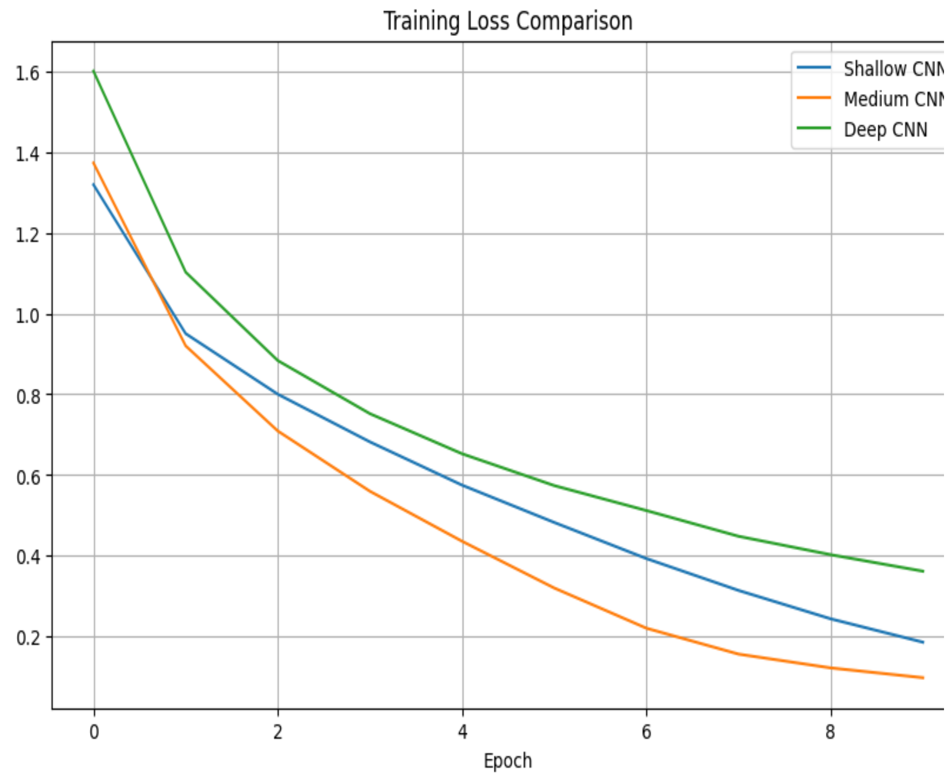


Training Loss Comparison

- All CNN models showed a consistent decrease in training loss, indicating effective learning and optimization.
- The **Medium CNN achieved the lowest final loss (0.0985)**, demonstrating the most efficient learning process.
- The **Shallow CNN reached a final loss of 0.1864**, but its limited depth restricted its ability to learn complex features.
- The **Deep CNN had the highest final loss (0.3621)**, as its greater complexity and dropout regularization resulted in slower convergence during training.

Key Finding

- The Medium CNN achieved the lowest training loss, indicating the most effective optimization and learning performance among all models.

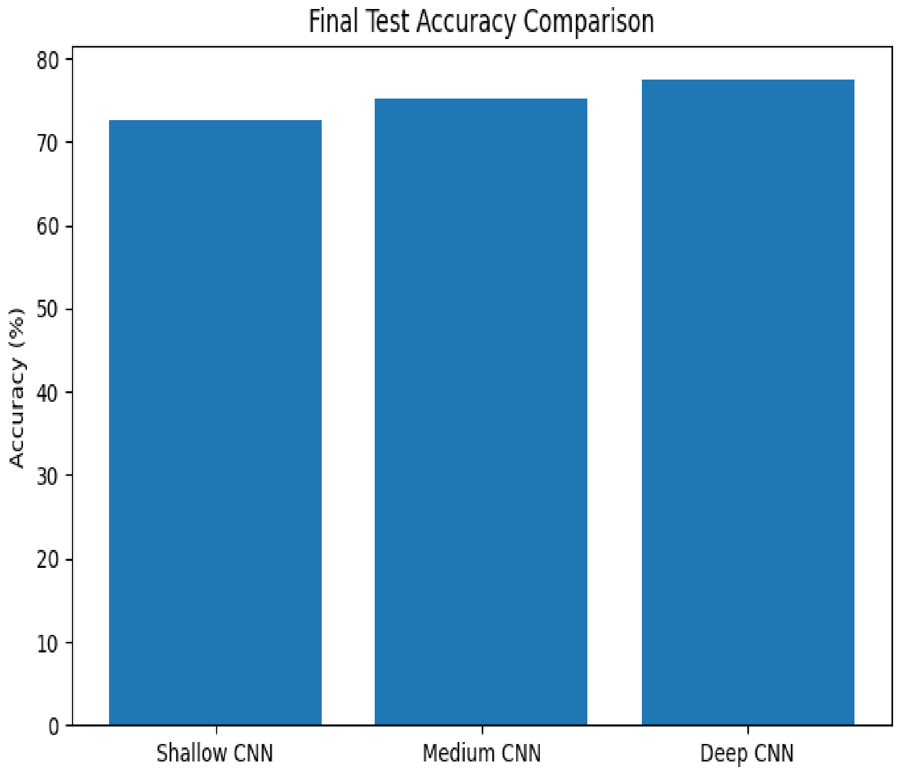


Performance Evaluation

- The **Deep CNN achieved the highest test accuracy (77.57%)**, demonstrating the strongest generalization performance on unseen data.
- The **Medium CNN achieved 75.14% accuracy**, performing well but learning slightly less complex features than the Deep CNN.
- The **Shallow CNN achieved 72.67% accuracy**, as its limited depth restricted its ability to capture detailed image patterns.

Key Finding

- The Deep CNN delivered the best overall performance, showing that deeper architectures can learn more discriminative features and improve image classification accuracy.

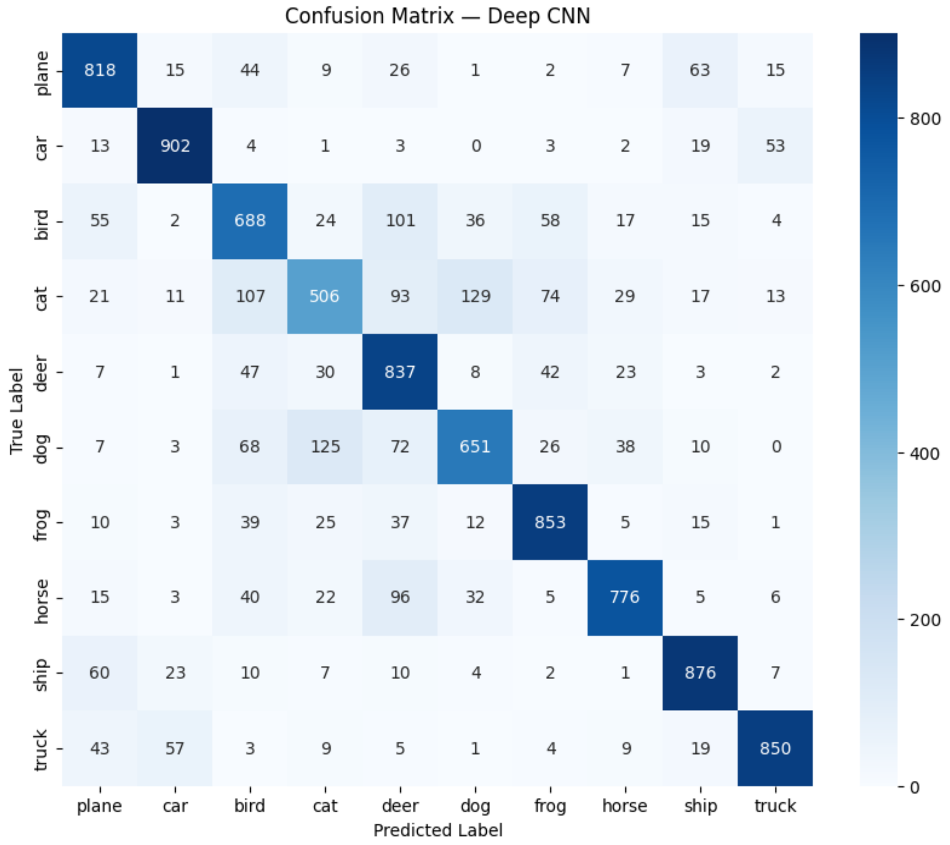


Confusion Matrix Analysis

- The confusion matrix illustrates the class-wise prediction performance of the **Deep CNN** on the CIFAR-10 test dataset.
- Classes such as **car (902)**, **ship (876)**, **frog (853)**, and **truck (850)** achieved the highest number of correct predictions.
- Classes such as **cat (506)** and **bird (688)** were more challenging to classify due to visual similarities with other categories.
- Common misclassifications occurred between **cat and dog**, and **deer and horse**, indicating overlapping visual features.

Key Finding

- The Deep CNN achieved strong overall classification performance, but visually similar classes remained the primary source of prediction errors.



Result Summary

- The **Deep CNN achieved the highest test accuracy (77.57%)**, followed by the **Medium CNN (75.14%)** and **Shallow CNN (72.67%)**.
- Although the **Medium CNN achieved the highest training accuracy**, the **Deep CNN generalized better to unseen data** due to its deeper architecture and dropout regularization.
- This demonstrates that **higher training accuracy does not always result in better real-world performance**.

Key Finding

- The Deep CNN delivered the best overall performance, achieving the highest test accuracy and strongest generalization capability. Although the Deep CNN achieved the highest test accuracy, training was limited to 10 epochs and no data augmentation was applied. Future work could investigate deeper architectures, data augmentation, and transfer learning.

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- PyTorch Documentation https://pytorch.org/docs/stable/index.html